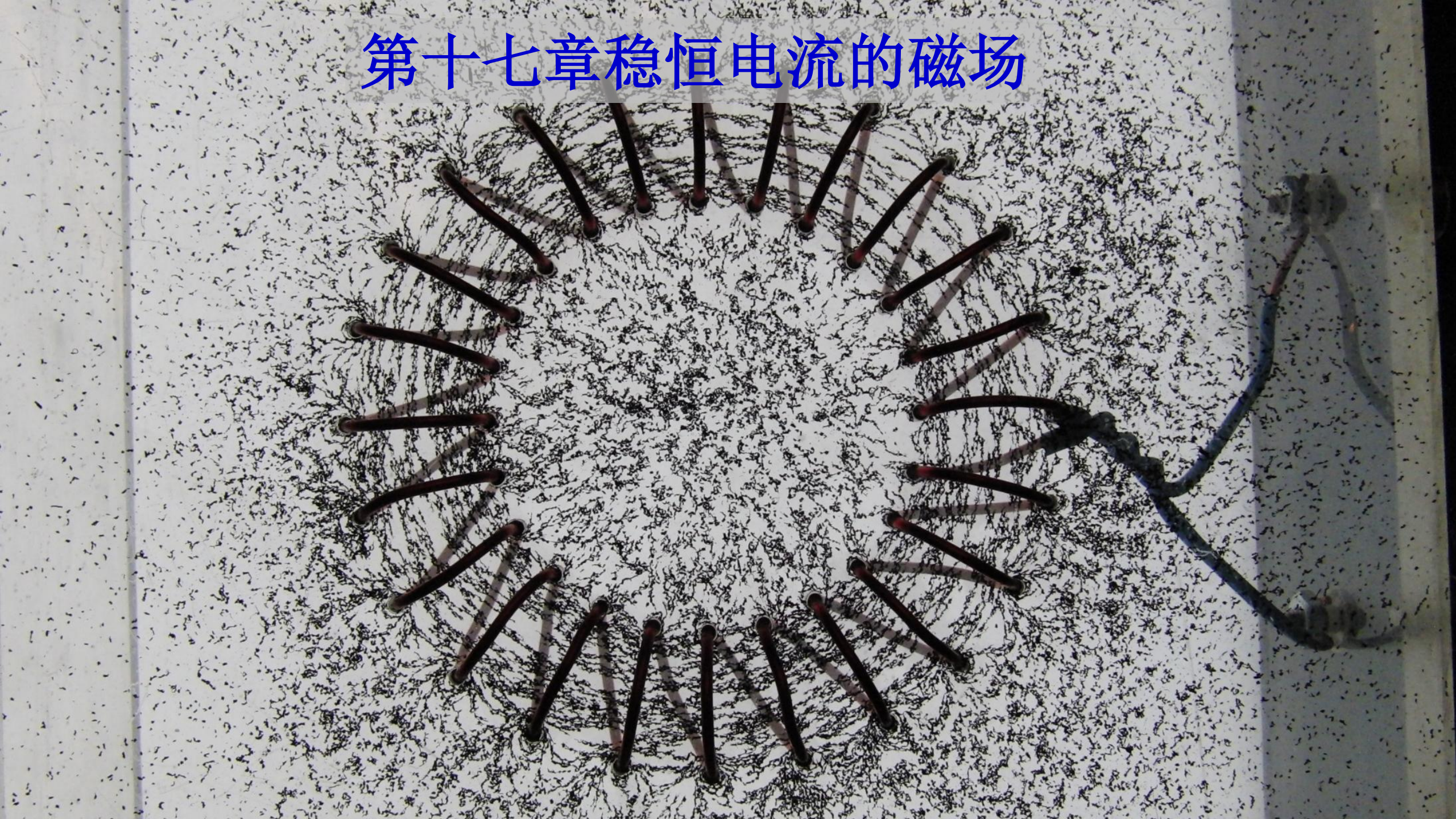
第十七章稳恒电流的磁场



- 核心概念
  - 磁感应强度
  - 磁偶极子
  - 磁矩
  - 磁通量
- 核心问题
  - 如何计算电流产生的磁场？
  - 磁场的高斯定理及其应用？
  - 安培环路定理及其应用？

第十七章（真空中）稳恒电流的磁场 (magnetic field)

17.1 基本磁现象

17.2 磁场　磁感强度

17.3 毕—萨—拉定律及应用

17.4 磁通量　磁场的高斯定理 磁力线

17.5 安培环路定理及应用

17.1 基本磁现象

磁石吸铁(2千年)　指南针

1820年 奥斯特　电流的磁效应　（演示实验）

法国物理学家

阿拉果　安培　毕奥 萨伐尔　拉普拉斯

从奥斯特发现到对磁现象的系统认识

只用半年时间

17.2 磁场　磁感强度

一. 磁场

电流产生磁场！　(运动电荷产生磁场?)

磁场对电流有力的作用　安培力

磁场对运动电荷的作用力：洛伦兹力　$\boxed{\vec{f} = q\vec{v} \times \vec{B}}$

二. 磁感应强度

利用运动电荷在磁场中受力定义磁场强度

$\vec{B}$　磁感强度

也称磁通密度 (magnetic flux density)

单位：特斯拉(T)=10^4高斯(G)

洛仑兹力公式

$$\boxed{\vec{f} = q\vec{E} + q\vec{v} \times \vec{B}}$$

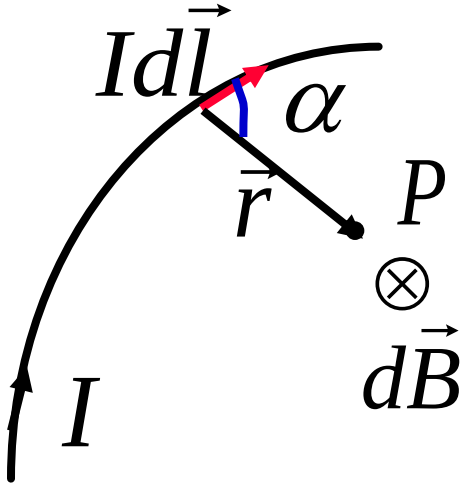
点源　　场　　统一

- 洛仑兹力是电荷受电磁场作用力的基本关系式
- 洛仑兹力是相对论不变式

17.3 毕—萨—拉定律及应用

一. 毕奥-萨伐尔-拉普拉斯定律

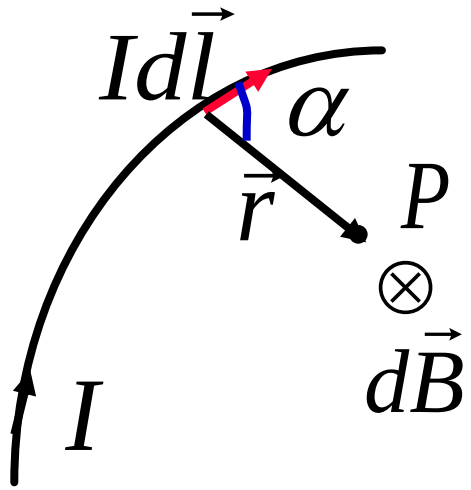
电流元 **(current element)** $Id\vec{l}$

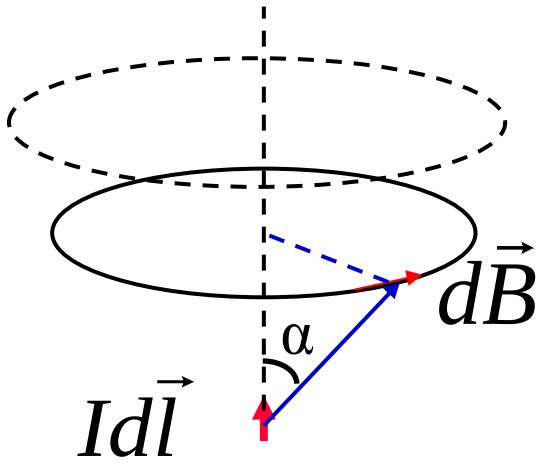


$$\boxed{d\vec{B} = \frac{\mu_0 I d\vec{l} \times \hat{r}}{4\pi r^2}}$$

μ_0 真空中的磁导率

$$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H/m}$$





$$d\vec{B} = \frac{\mu_0 I d\vec{l} \times \hat{r}}{4\pi r^2}$$

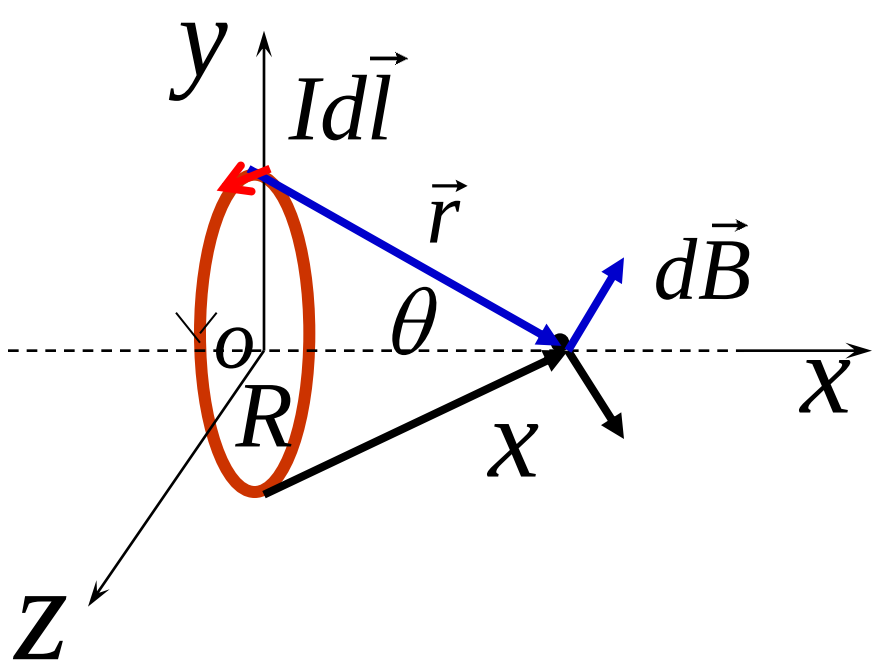
$$dB = \frac{\mu_0 I dl \sin\alpha}{4\pi r^2}$$

磁场叠加原理 $\vec{B} = \int d\vec{B}$

例1 求圆电流轴线上的磁感强度

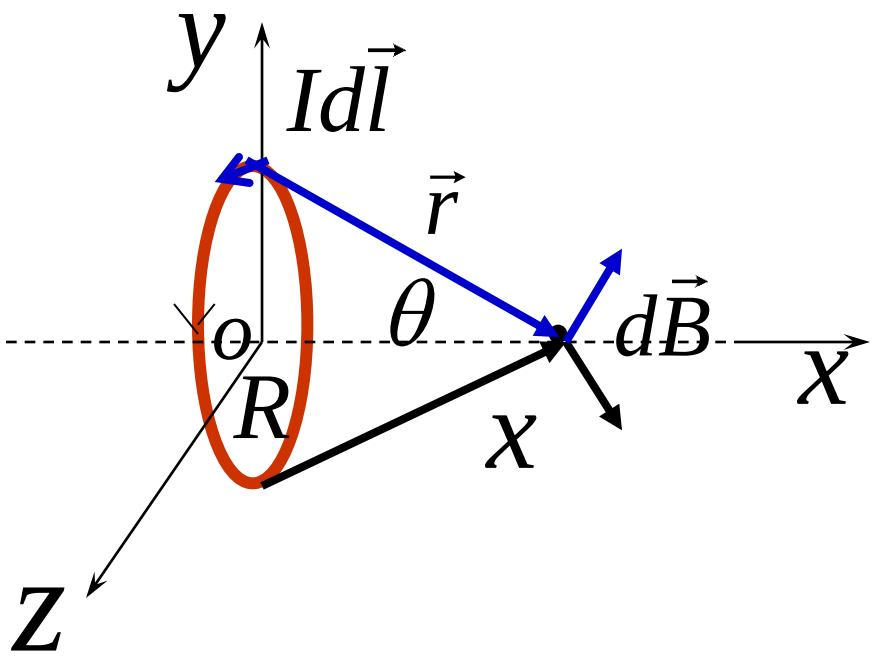
解：在圆环上任取电流元

$$d\vec{B} = \frac{\mu_0 I d\vec{l} \times \hat{r}}{4\pi r^2}$$



由对称性垂直x 轴的场强为 0

$$\vec{B} = B_x \hat{x} \qquad dB_x = dB\sin\theta = \frac{\mu_0}{4\pi}\frac{Idl}{r^2}\frac{R}{r}$$



$$B_x = \int \frac{\mu_0}{4\pi} \frac{Idl}{r^2} \frac{R}{r}$$

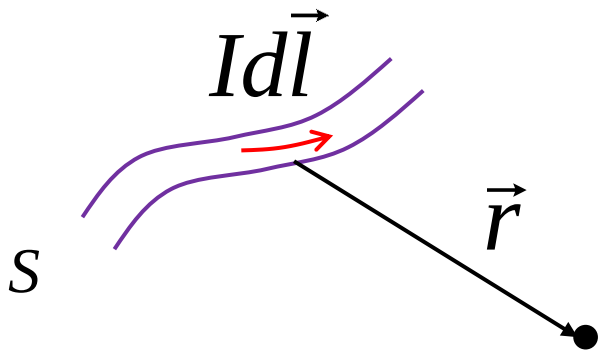
$$= \frac{\mu_0}{2} \frac{IR^2}{r^3}$$

$$B = \frac{\mu_0}{2} \frac{IR^2}{(R^2 + x^2)^{3/2}}$$

方向: 沿轴线右手螺旋

圆心处: $B = \dfrac{\mu_0 I}{2R}$

例2 匀速运动点电荷的磁场



$$d\vec{B} = \frac{\mu_0 Id\vec{l} \times \hat{r}}{4\pi r^2}$$

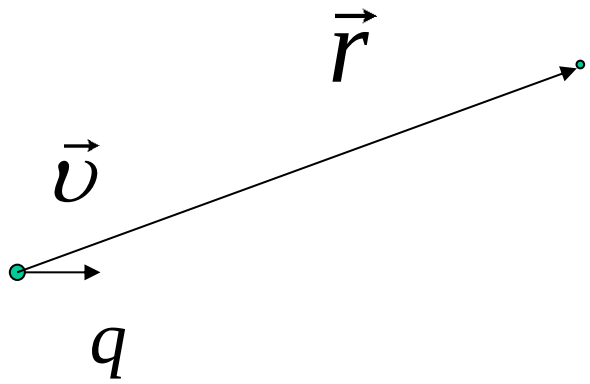
$$\frac{d\vec{B}}{dN}$$

$$Id\vec{l} = S\vec{j}dl$$

$$= nq\vec{v}dV$$

$$= q\vec{v}dN$$

$$\boxed{\vec{B} = \frac{\mu_0 q\vec{v} \times \hat{r}}{4\pi r^2}}$$



$$\vec{B} = \frac{\mu_0 q \vec{\upsilon} \times \hat{r}}{4\pi r^2}$$

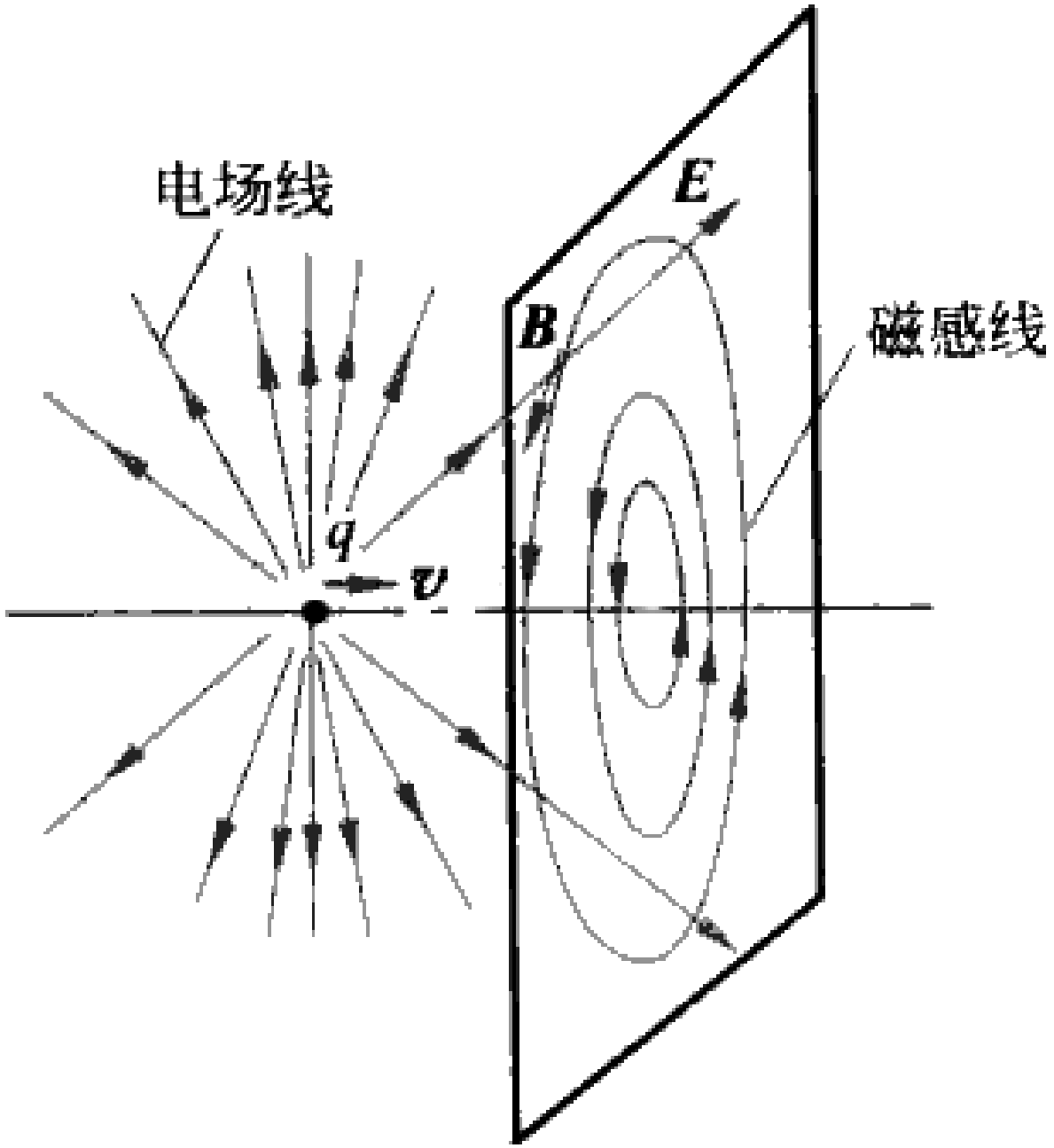
低速近似

$$\vec{E} = \frac{q\hat{r}}{4\pi\varepsilon_0 r^2}$$

$$c^2 = \frac{1}{\varepsilon_0 \mu_0}$$

$$\boxed{\vec{B} = \frac{1}{c^2}\vec{v} \times \vec{E}}$$

准确



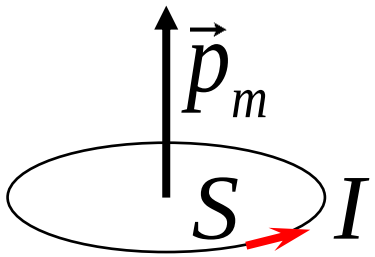
低速近似

$$\vec{B} = \frac{\mu_0 q \vec{v} \times \hat{r}}{4\pi r^2}$$

$$\vec{E} = \frac{q\hat{r}}{4\pi\varepsilon_0 r^2}$$

$$\vec{B} = \frac{1}{c^2}\vec{v} \times \vec{E}$$

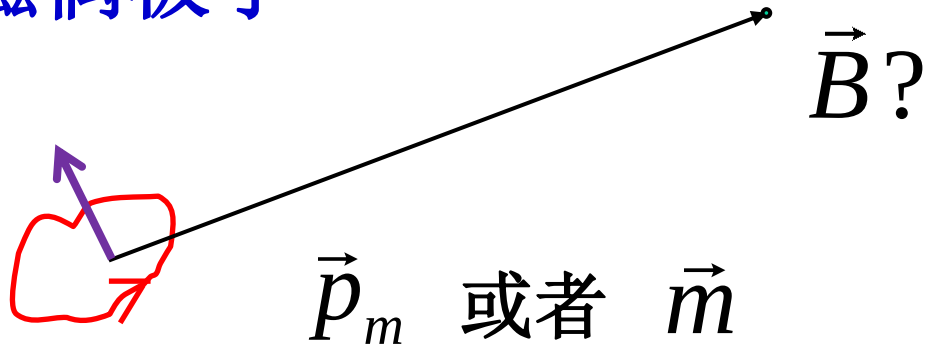
载流线圈的磁矩及其磁场



$$\boxed{\vec{p}_m = I\vec{S}}$$

平面载流线圈

磁偶极子



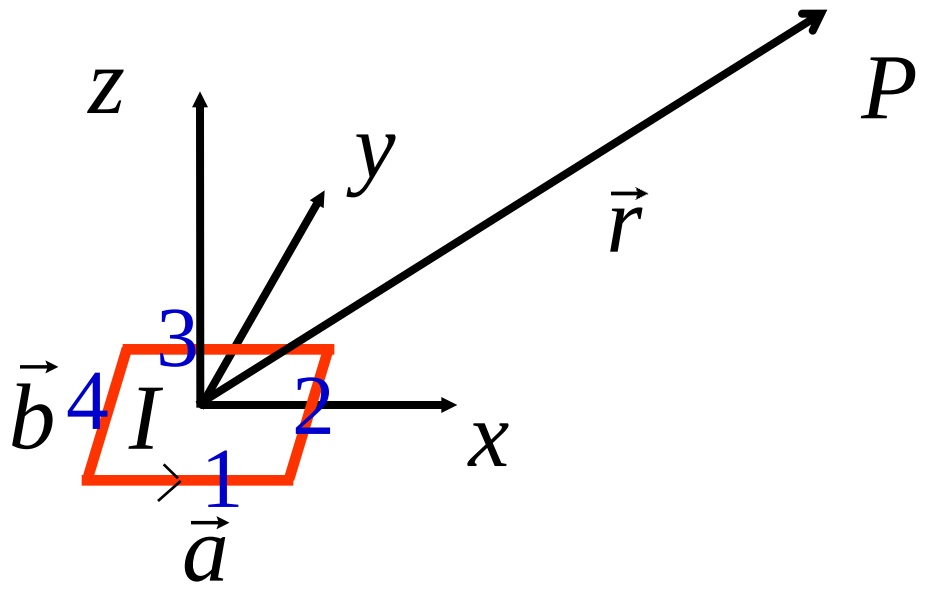
场点距平面线圈的距离 $r \gg \bar{d}$

定义平面载流线圈的磁矩

[magnetic (dipole) moment]

非平面线圈也有磁矩

$$\vec{B} = \frac{\mu_0}{4\pi}\int \frac{I d\vec{l} \times \hat{r}}{r^2}$$



小矩形电流在 **z = 0**

磁矩

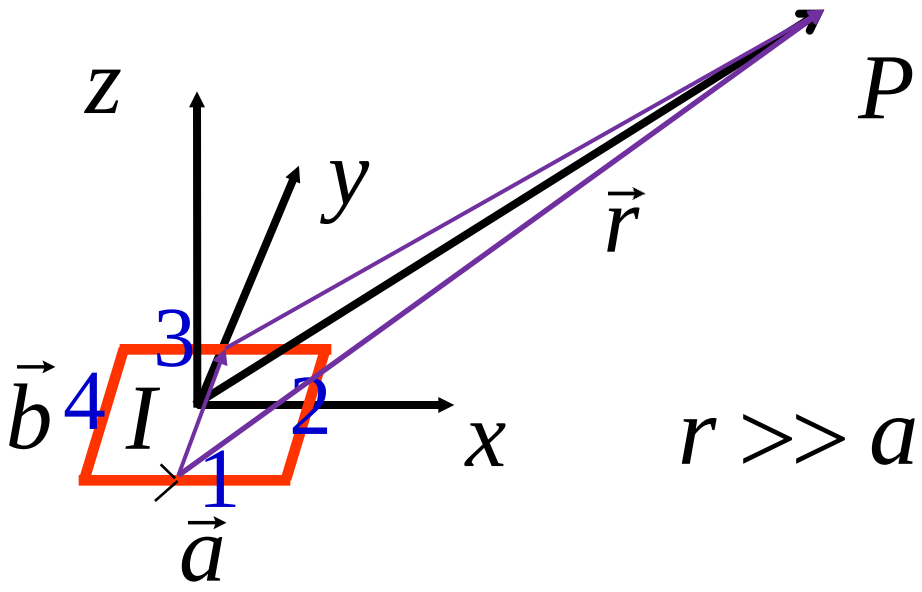
$$\vec{p}_m = Iab\hat{z}$$

每个边当作一个电流元

$$r >> a, \quad r >> b$$

$$\vec{B} = \frac{\mu_0 I}{4\pi}\left(\frac{\vec{a}\times\vec{r}_1}{r_1^3} - \frac{\vec{a}\times\vec{r}_3}{r_3^3} + \frac{\vec{b}\times\vec{r}_2}{r_2^3} - \frac{\vec{b}\times\vec{r}_4}{r_4^3}\right)$$

小矩形电流在 z = 0



$$r >> a, \quad r >> b$$

$$\vec{r}_1 = \vec{r} + \frac{\vec{b}}{2} \qquad \vec{r}_3 = \vec{r} - \frac{\vec{b}}{2}$$

$$\vec{r}_2 = \vec{r} - \frac{\vec{a}}{2} \qquad \vec{r}_4 = \vec{r} + \frac{\vec{a}}{2}$$

$$\vec{r}_1 = \vec{r} + \frac{\vec{b}}{2} \qquad \vec{r}_1^{\,2} = \vec{r}^{\,2} + \frac{\vec{b}^{\,2}}{4} + \vec{r}\cdot\vec{b}$$

$$(1+x)^{\alpha} \approx 1 + \alpha x \qquad \frac{1}{r_1^3} \approx \frac{1}{r^3}\left(1 - \frac{3}{2}\frac{\vec{r}\cdot\vec{b}}{r^2}\right)$$

$$(\vec{A}\times\vec{B})\times\vec{C} = \vec{B}(\vec{A}\cdot\vec{C}) - \vec{A}(\vec{B}\cdot\vec{C}) \qquad \vec{p}_m = Iab\hat{z}$$

$$\vec{B} = \frac{\mu_0}{4\pi r^3}\left[-\vec{p}_m + 3\left(\vec{p}_m\cdot\hat{r}\right)\hat{r}\right]$$

结果与磁矩的具体几何形状无关

比较电偶极子电场

$$\vec{E} = \frac{1}{4\pi\varepsilon_0 r^3}\left[-\vec{p}_e + 3\left(\vec{p}_e\cdot\hat{r}\right)\hat{r}\right]$$

*推导说明：$\vec{B}=\frac{\mu_0 I}{4\pi}(\frac{\vec{a}\times\vec{r}_1}{r_1^3}-\frac{\vec{a}\times\vec{r}_3}{r_3^3}+\frac{\vec{b}\times\vec{r}_2}{r_2^3}-\frac{\vec{b}\times\vec{r}_4}{r_4^3})$

$$\vec{B}=\frac{\mu_0 I}{4\pi}(\frac{\vec{a}\times(\vec{r}+\vec{b}/2)(1-\frac{3}{2}\frac{\vec{r}\cdot\vec{b}}{r^2})}{r^3}-\frac{\vec{a}\times(\vec{r}-\vec{b}/2)(1+\frac{3}{2}\frac{\vec{r}\cdot\vec{b}}{r^2})}{r^3}$$

$$+\frac{\vec{b}\times(\vec{r}-\vec{a}/2)(1+\frac{3}{2}\frac{\vec{r}\cdot\vec{a}}{r^2})}{r^3}-\frac{\vec{b}\times(\vec{r}+\vec{a}/2)(1-\frac{3}{2}\frac{\vec{r}\cdot\vec{a}}{r^2})}{r^3})$$

$$\vec{B}=\frac{\mu_0 I}{4\pi}(\frac{2\vec{a}\times\vec{b}}{r^3}-\frac{3\vec{a}\times\vec{r}(\vec{r}\cdot\vec{b})}{r^5}+\frac{3\vec{b}\times\vec{r}(\vec{r}\cdot\vec{a})}{r^5})$$

$$\vec{B}=\frac{\mu_0 I}{4\pi}(\frac{2\vec{S}}{r^3}-\frac{3[\vec{a}(\vec{r}\cdot\vec{b})-\vec{b}(\vec{r}\cdot\vec{a})]\times\vec{r}}{r^5})$$

利用 $(\vec{A}\times\vec{B})\times\vec{C}=\vec{B}(\vec{A}\cdot\vec{C})-\vec{A}(\vec{B}\cdot\vec{C})$

$$\vec{B}=\frac{\mu_0 I}{4\pi}(\frac{2\vec{S}}{r^3}-\frac{3[(\vec{b}\times\vec{a})\times\vec{r}]\times\vec{r}}{r^5})$$

$$\vec{B}=\frac{\mu_0 I}{4\pi}(\frac{2\vec{S}}{r^3}+\frac{3(\vec{S}\times\vec{r})\times\vec{r}}{r^5})$$

$$\vec{B}=\frac{\mu_0 I}{4\pi}(\frac{2\vec{S}}{r^3}+\frac{3[\vec{r}(\vec{S}\cdot\vec{r})-\vec{S}(\vec{r}\cdot\vec{r})]}{r^5})$$

$$\vec{B}=\frac{\mu_0}{4\pi}(-\frac{I\vec{S}}{r^3}+\frac{3\vec{r}(I\vec{S}\cdot\vec{r})}{r^5})$$

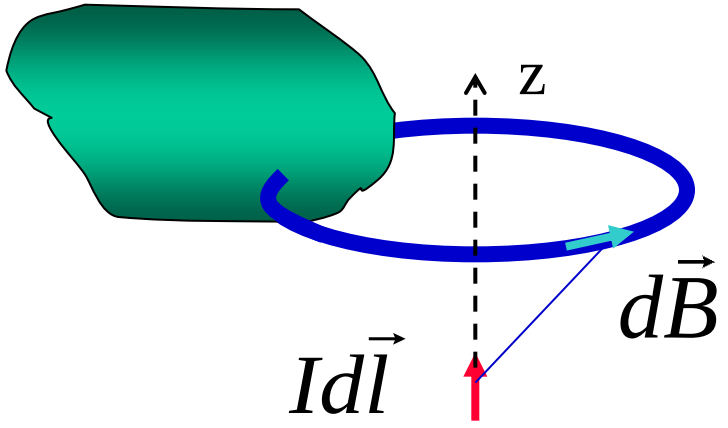
$$\vec{B}=\frac{\mu_0}{4\pi}(-\frac{\vec{p}_m}{r^3}+\frac{3\vec{r}(\vec{p}_m\cdot\vec{r})}{r^5})$$

17.4 磁场的高斯定理

一. 磁通量

$$\phi_m = \iint_S \vec{B} \cdot d\vec{S}$$ 单位：韦伯(Wb)

二. 磁通连续原理（磁场的高斯定理）



电流元磁通量为零

根据叠加原理

$$\boxed{\oiint_S \vec{B} \cdot d\vec{S} = 0}$$

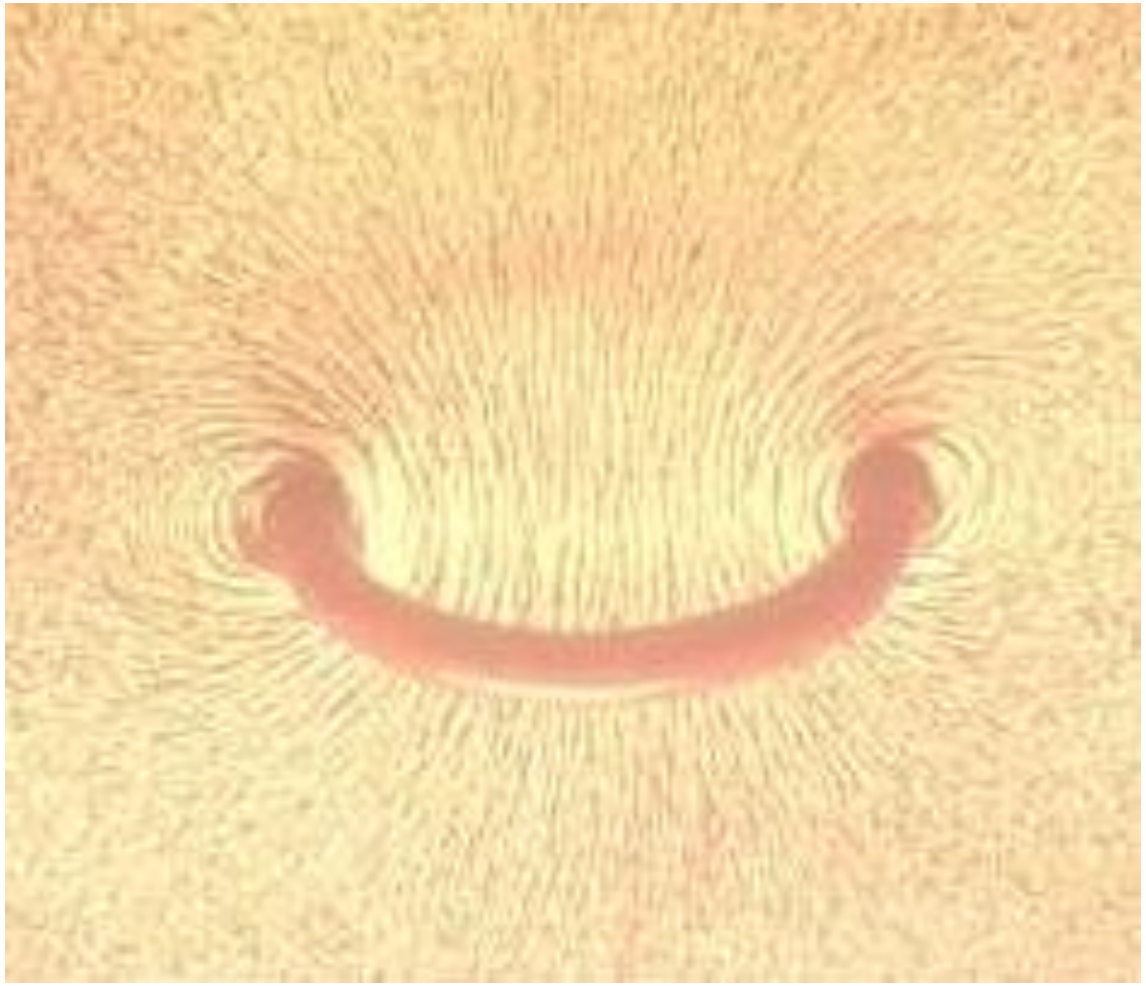
无源场 = 没有磁荷

三. 磁场线（磁力线）

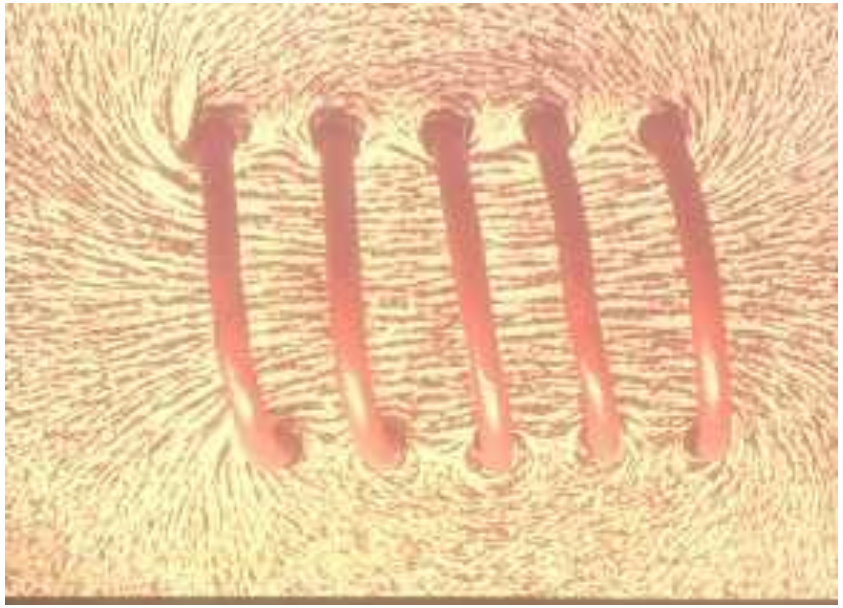
1. 典型电流的磁力线

各种典型的磁感应线的分布：

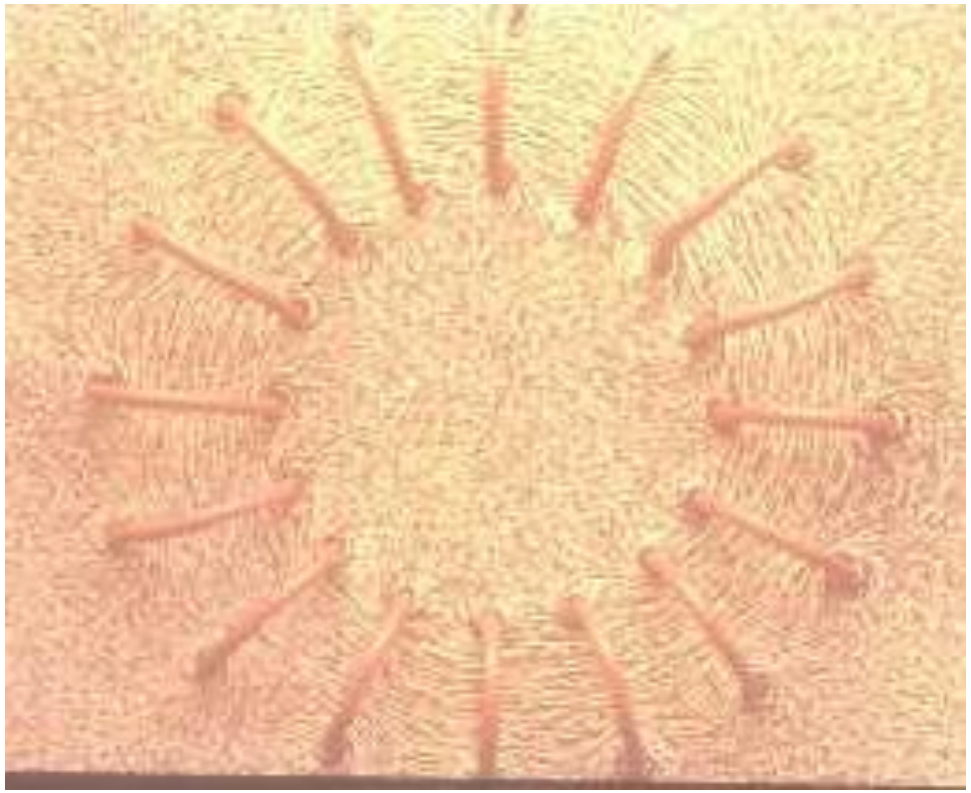
直线电流的磁感线



圆形电流的磁感线



直螺线管电流的磁感线



环形螺线管电流的磁感线

2. 磁力线的性质

- 无头无尾闭合曲线
- 与电流套连
- 与电流成右手螺旋关系

*磁单极（magnetic monopole）：

根据电和磁的对称性：

$$\oiint_S \vec{D}\cdot\overrightarrow{\mathrm{d}S} = q_0 \longrightarrow \oiint_S \vec{B}\cdot\overrightarrow{\mathrm{d}S} = q_m \qquad \boldsymbol{q_m} \text{ — 磁荷}$$

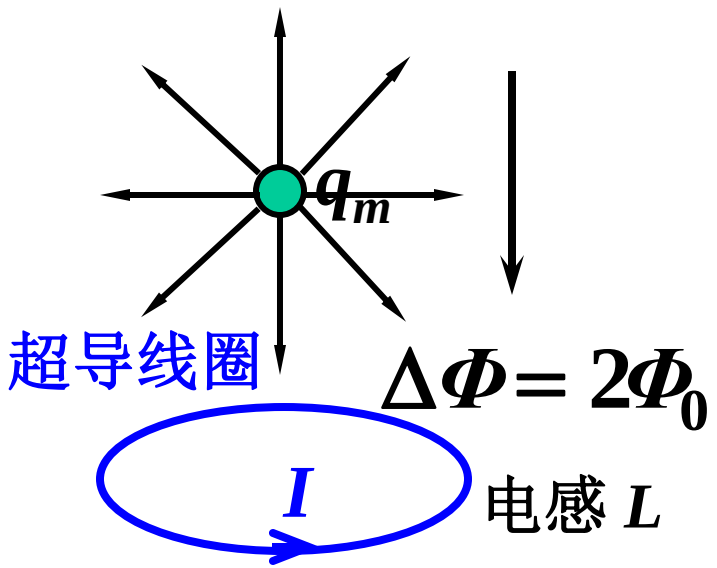
1931，Dirac预言了磁单极子的存在。

$$\boldsymbol{q\cdot q_m = nh,\ (n=1,\ 2,\ 3\cdots)},\quad \boldsymbol{h}\text{是普朗克常量}$$

只要存在磁单极子就能证明电荷的量子化。

磁单极子质量： $m = 2\times10^{-11}\,\mathrm{g} \approx 10^{16} m_{\mathrm{p}}$

希望从宇宙射线中捕捉到磁单极子



有磁单极子穿过时，感应电流

$$I = 2\Phi_0 / L$$

斯坦福大学Cabrera等人记录到一次电流的跃变(1982)

目前仍然不能在实验中确认磁单极子存在。

17.5 安培环路定理及应用

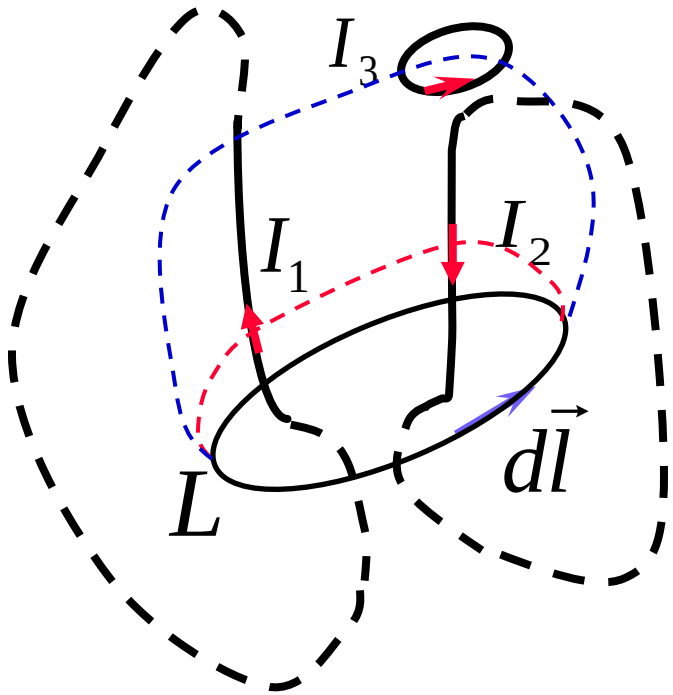
一. 定理表述

在恒定磁场中，磁感强度 $\vec{B}$

沿任一闭合环路的线积分，等于穿

过该环路的所有电流的代数和的 μ_0倍。

$$\boxed{\oint_L \vec{B}\cdot d\vec{l} = \mu_0 \sum_i I_{i内}}$$



$$\oint_L \vec{B}\cdot d\vec{l} = \mu_0 \sum_i I_{i内}$$

$I_1 - I_2$

任一 闭合线L的绕行方向给定

磁场的环路积分等于穿过L为边界的某曲面的电流代数和的μ_0倍。

与L绕行方向成右手螺旋电流取正

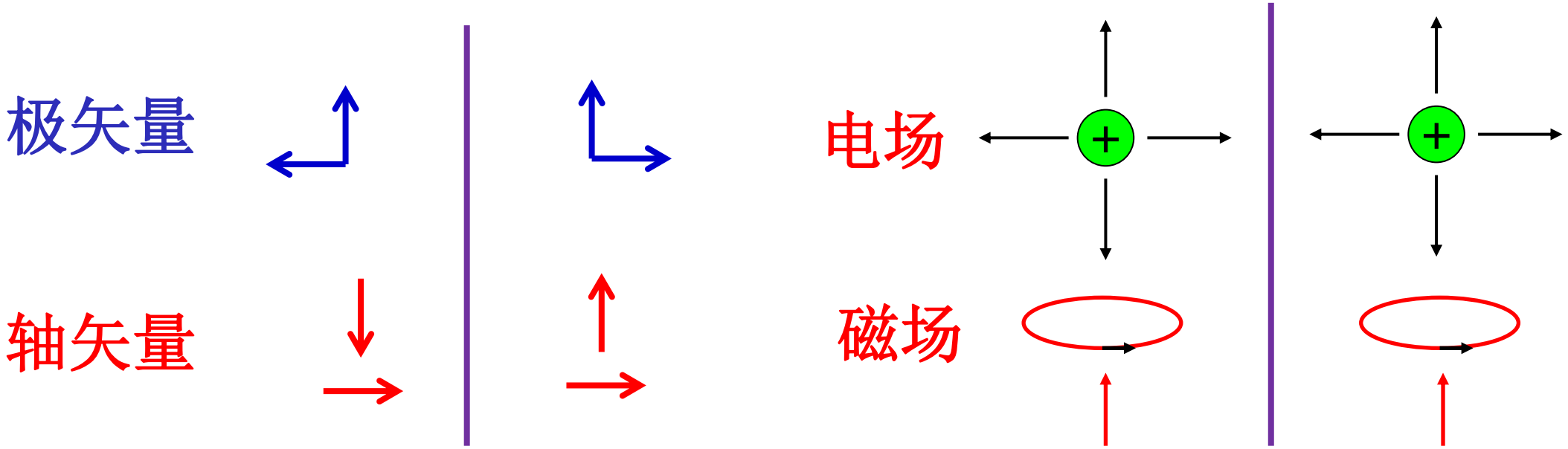
- 环路积分不为零, 不能定义标量势

二. 安培环路定理的应用

对于一些对称分布的电流，可以利用磁场的环路定理比较方便地求解场量

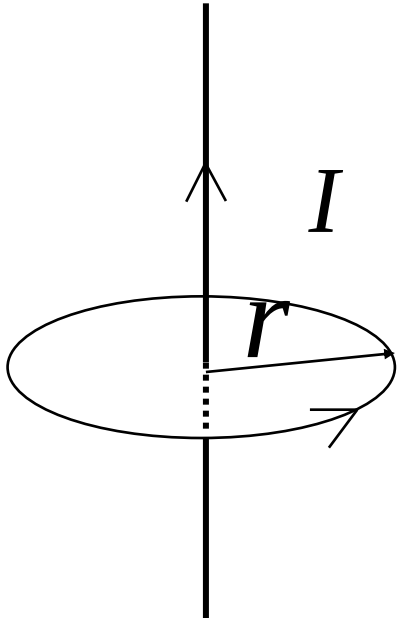
磁场由电流产生, 电流分布有对称性,
相应的磁场亦有对称性(不变性)

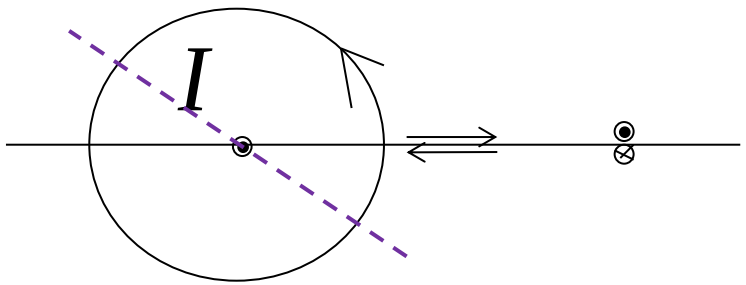
镜像对称性



例4 求长直导线电流周围的磁感强度

根据电流的轴面镜像对称性　磁场是轴矢量





有很多镜像对称面

在镜像对称面上不可能有平行于对称面的磁场分量。

根据电流的轴对称性　磁场在圆环上大小相同　同心环磁场

$$\oint_L \vec{B}\cdot d\vec{l} = \oint_L B dl = B2\pi r = \mu_0 I \qquad B = \frac{\mu_0}{2\pi}\frac{I}{r}$$

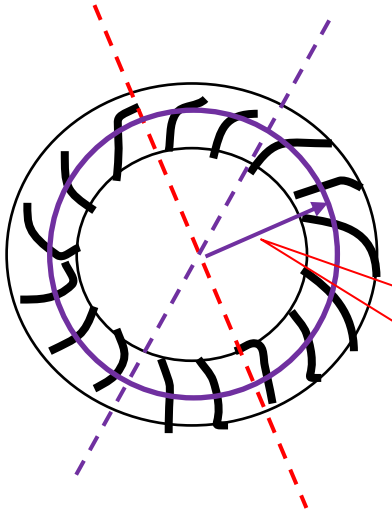
由安培环路定理可解一些典型的场

密绕螺绕环

镜像对称面

匝数

$$B2\pi r = \mu_0 NI$$

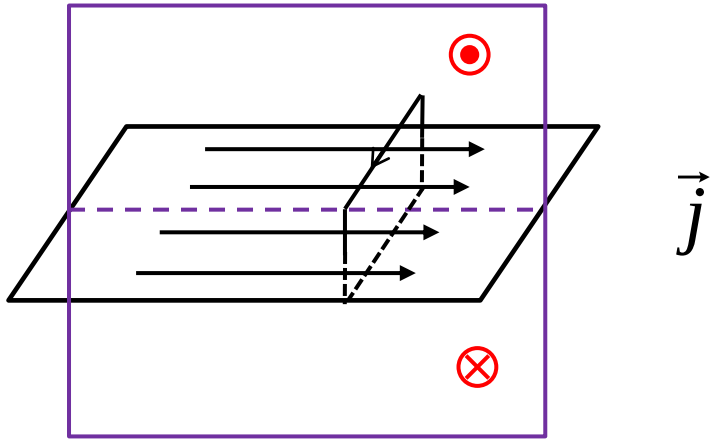


场点距中心的距离r

$$\boxed{B = \frac{\mu_0 NI}{2\pi r}}$$

同心环磁场

无限大均匀载流平面



$$\oint_L \vec{B}\cdot d\vec{l} = 2Bl = \mu_0 l j$$

j是面电流密度

$$\boxed{B = \frac{\mu_0 j}{2}}$$

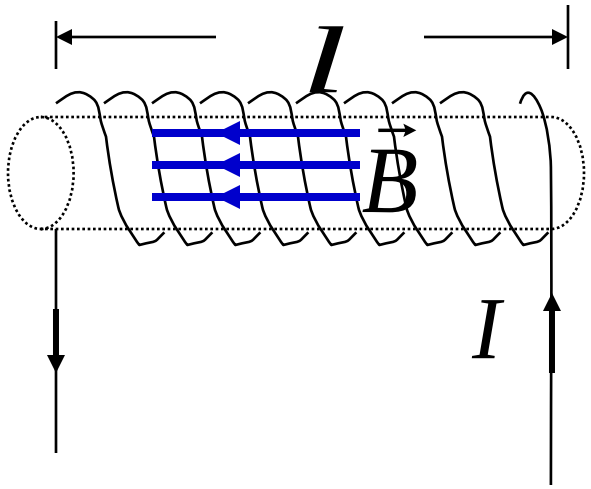
例5 求密绕长直螺线管内部的磁感强度

总匝数为N　总长为l

通过稳恒电流　电流强度为 I

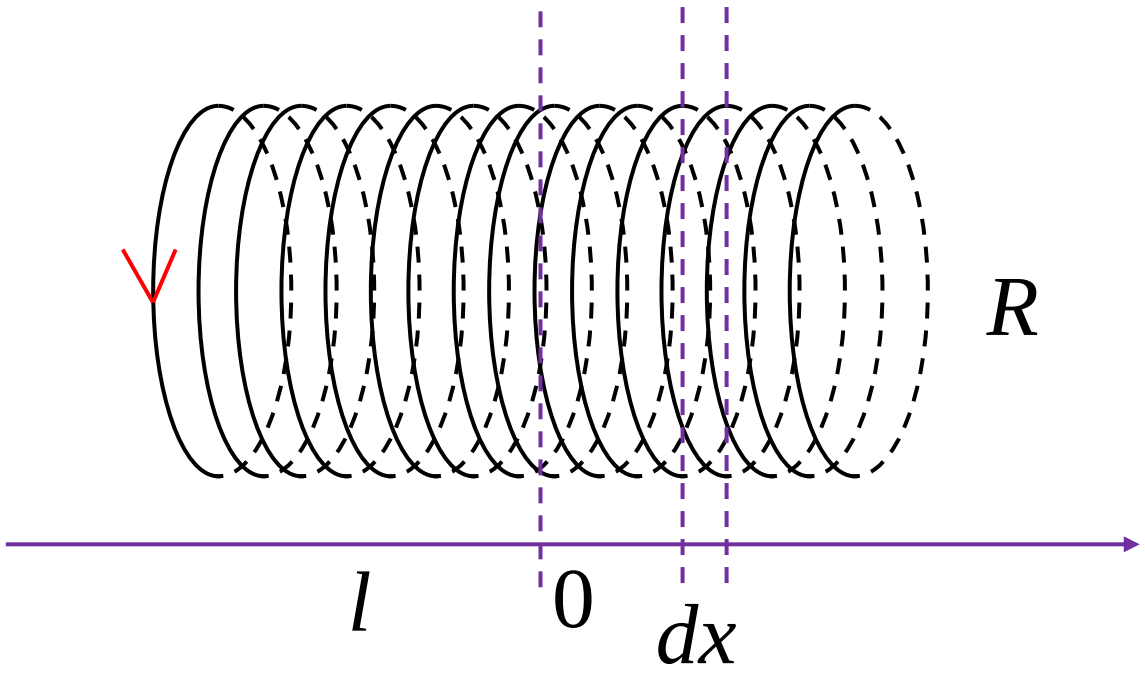
分析对称性 知内部场沿轴向

方向与电流成右手螺旋关系

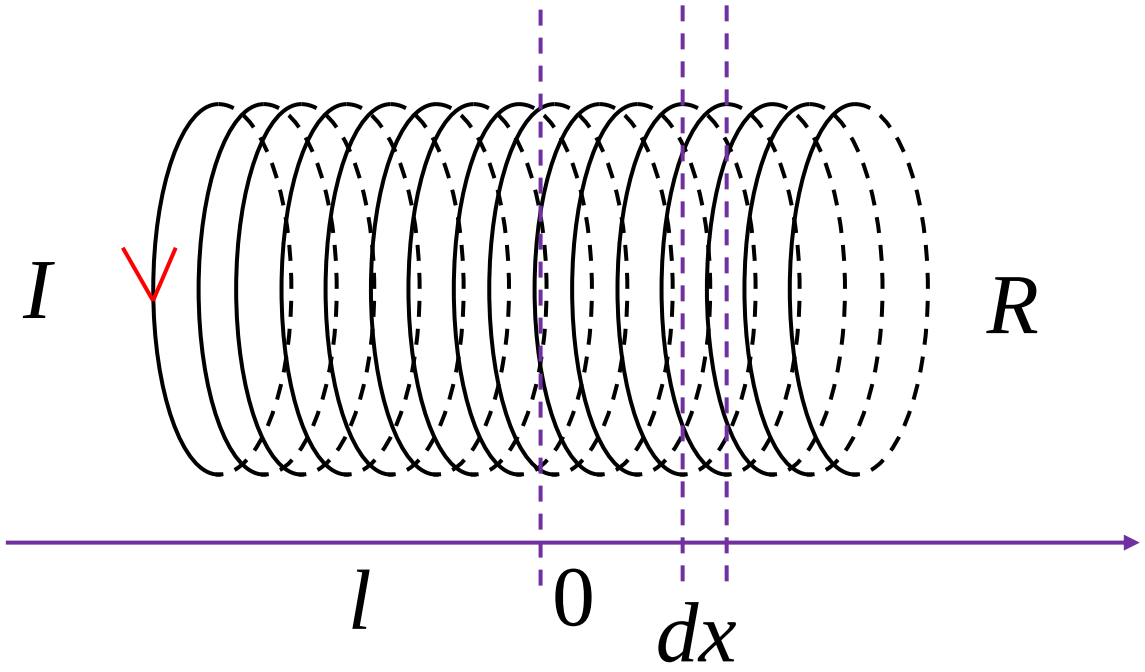


解：先利用叠加原理计算

密绕螺线管中心处磁感应强度



$$B=\frac{\mu_0 I}{2}\frac{R^2}{(R^2+x^2)^{3/2}} \qquad dB=\frac{\mu_0 I}{2}\frac{R^2}{(R^2+x^2)^{3/2}}\frac{N}{l}dx$$

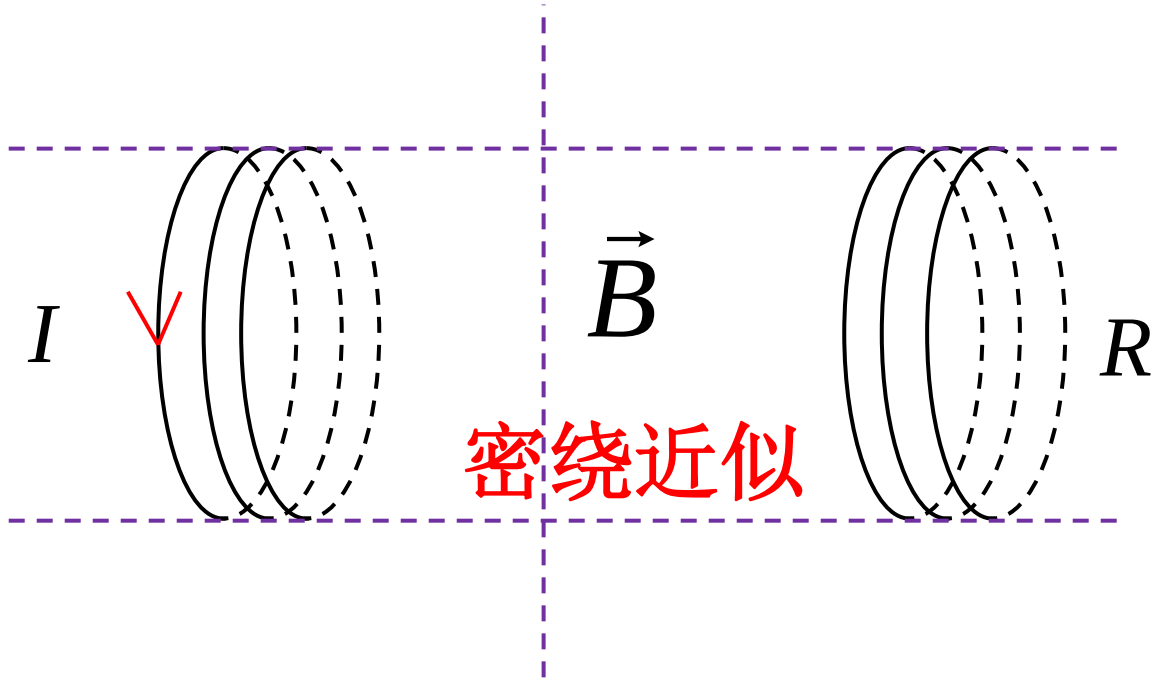


$$B=\frac{\mu_0 I}{2}\frac{N}{l}\int_{-l/2}^{l/2}\frac{R^2}{(R^2+x^2)^{3/2}}dx$$

$$=\mu_0\frac{N}{l}I\frac{l}{\sqrt{4R^2+l^2}}$$

无限长螺线管

$$\boxed{B=\mu_0 nI}$$



镜像对称面 磁场是轴矢量

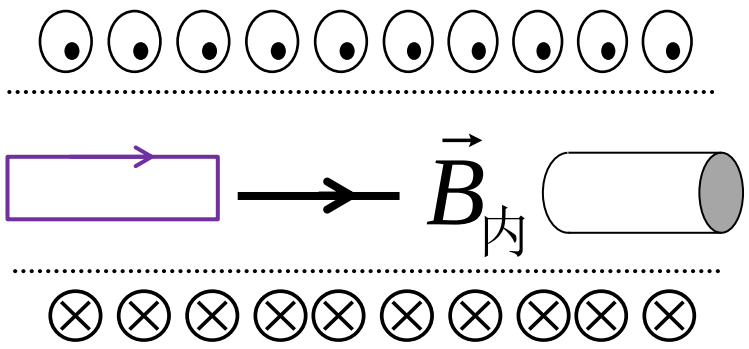
镜像对称面上不能有平行对称面的分量

只有垂直于对称面的分量

管内管外一样，磁场沿螺线管方向

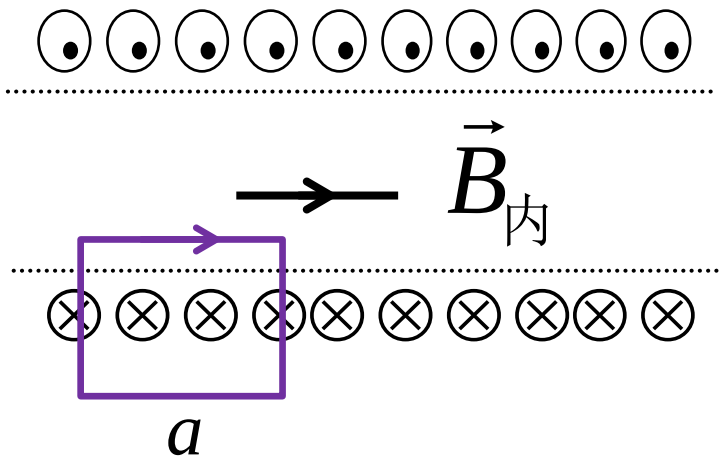
做一圆柱高斯面

做矩形环路



管内管外都是均匀场

密绕长直螺线管中 $B = \mu_0 nI$



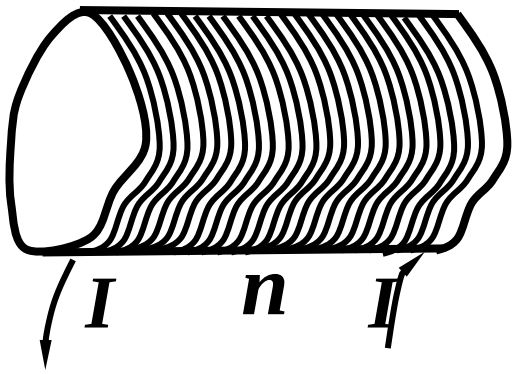
$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 naI$$

$$\parallel$$

$$aB_{内} + aB_{外}$$

$$B_{外} = 0$$

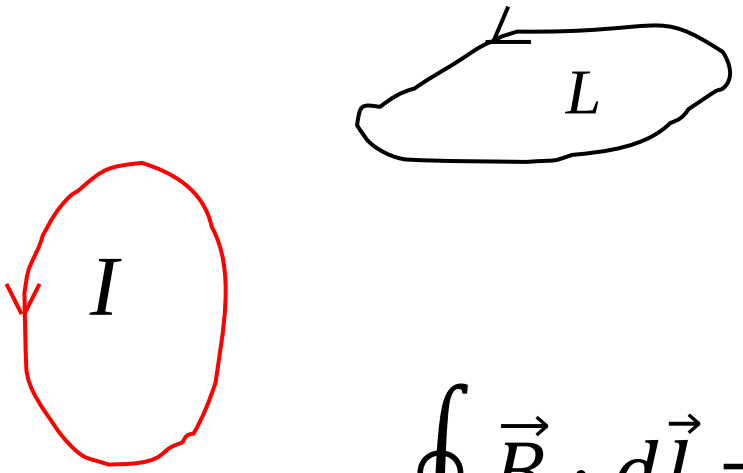
思考 截面形状任意的密绕长直螺线管内外的磁场如何？



提示：利用对称性（轴矢量、极矢量）

安培环路定理证明**

先考虑一个电回路的情况



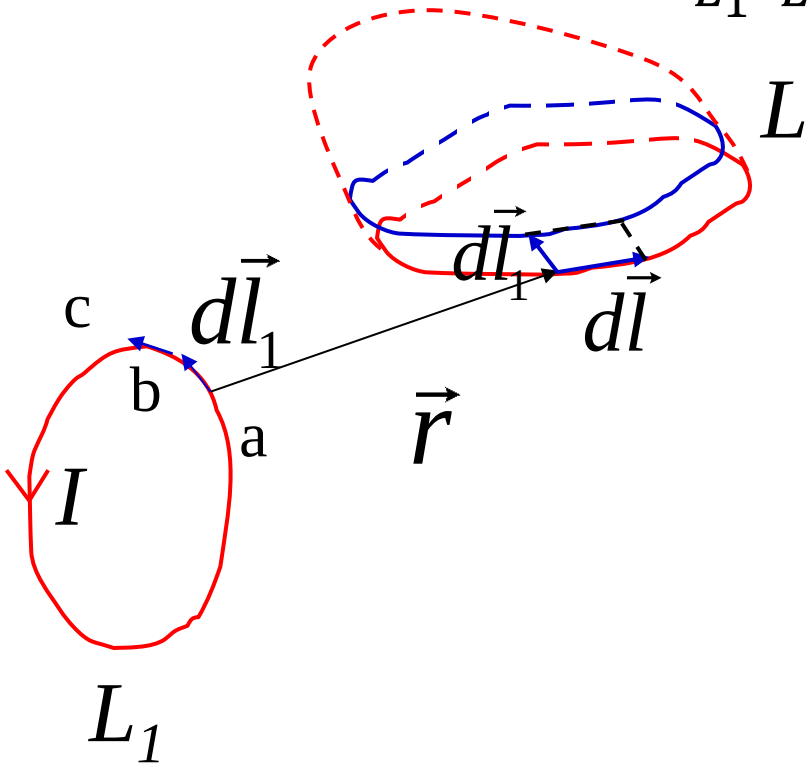
$$\oint_L \vec{B} \cdot d\vec{l} = \begin{cases} 0 \\ \pm\mu_0 I \end{cases}$$

证明

$$\oint_L \vec{B}\cdot d\vec{l} = \oint_L \oint_{L_1} \frac{\mu_0 I d\vec{l}_1 \times \hat{r}}{4\pi r^2}\cdot d\vec{l}$$

$$(\vec{A}\times\vec{B})\cdot\vec{C} = \vec{A}\cdot(\vec{B}\times\vec{C})$$

$$= \frac{\mu_0 I}{4\pi}\oint_{L_1}\oint_L \frac{d\vec{l}\times d\vec{l}_1}{r^2}\cdot\hat{r} \quad = \frac{\mu_0 I}{4\pi}\oint_{L_1} d\Omega$$



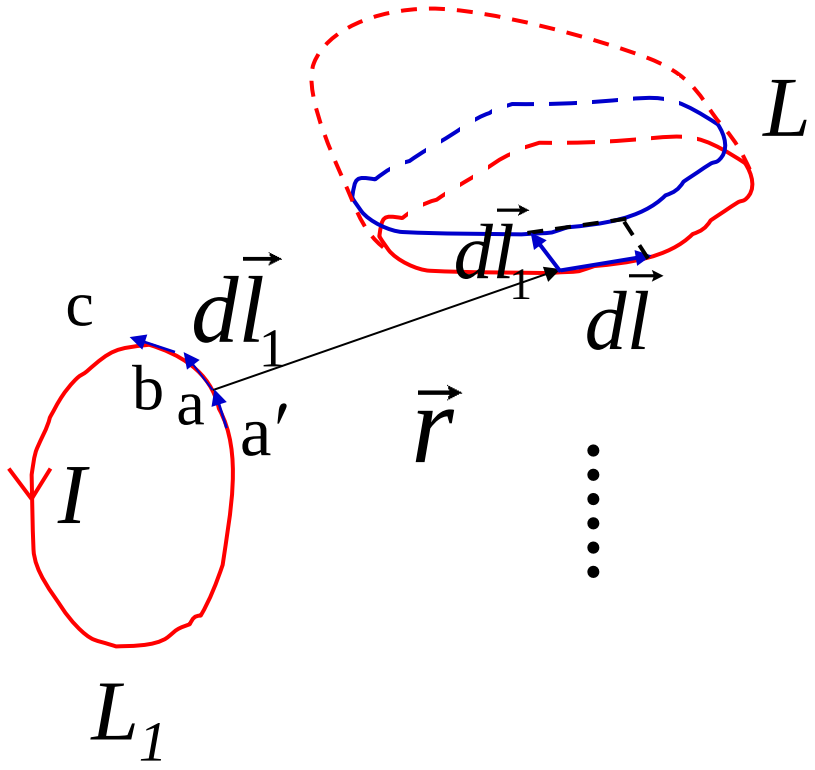
$$d\Omega_a = \Omega_{a红} - \Omega_{a蓝}$$

$$d\Omega_b = \Omega_{b红} - \Omega_{b蓝}$$

$$d\Omega_c = \Omega_{c红} - \Omega_{c蓝}$$

$$\vdots$$

$$\oint_L \vec{B}\cdot d\vec{l} = \frac{\mu_0 I}{4\pi}\oint_L d\Omega$$



$$d\Omega_a = \cancel{\Omega_{a红}} - \Omega_{a蓝}$$

$$d\Omega_b = \cancel{\Omega_{b红}} - \cancel{\Omega_{b蓝}}$$

$$d\Omega_c = \cancel{\Omega_{c红}} - \cancel{\Omega_{c蓝}}$$

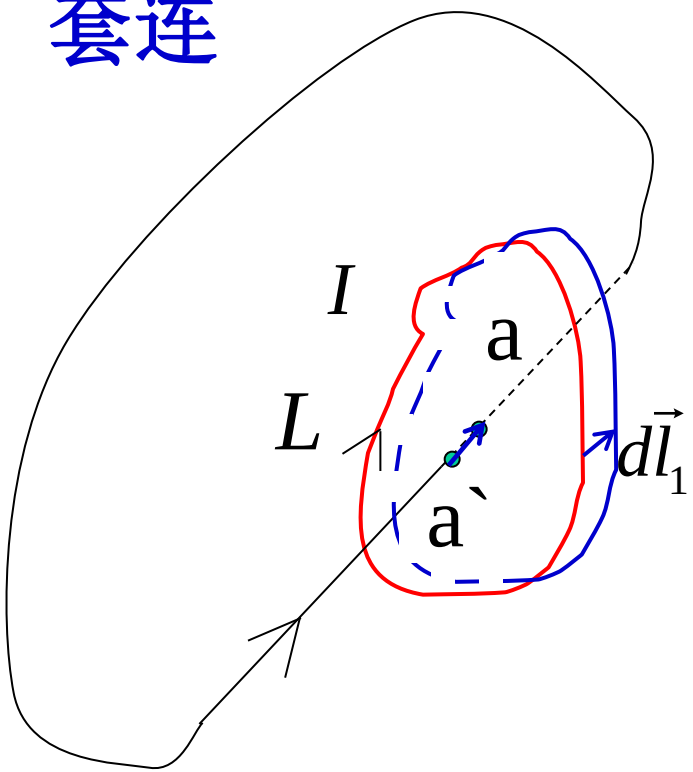
$$\vdots$$

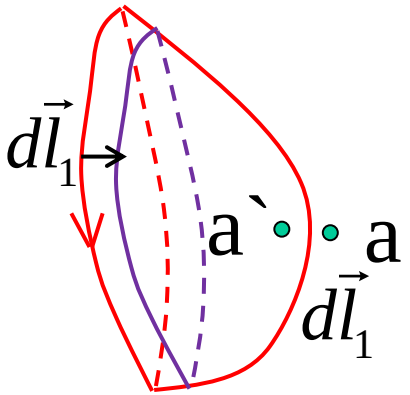
不套连

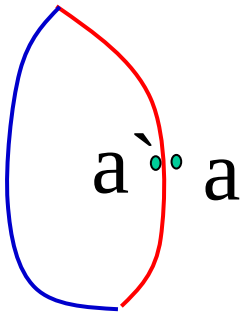
$$\oint_L d\Omega = \sum_i d\Omega_i = \Omega_{a'红} - \Omega_{a蓝} = 0$$

套连

$$\oint_L d\Omega = \Omega_{a'红} - \Omega_{a蓝}$$
$$= \Omega_{a'红} - \Omega_{a红} = 4\pi$$







$$\oint_L \vec{B} \cdot d\vec{l} = \frac{\mu_0 I}{4\pi} \oint_L d\Omega \quad = \mu_0 I$$

积分路径反向出负号

多个电流环叠加原理

$$\oint_L \vec{B}\cdot d\vec{l} = \oint_L \sum_i \vec{B}_i \cdot d\vec{l}$$

$$= \sum_i \oint_L \vec{B}_i \cdot d\vec{l}$$

$$= \mu_0 \sum_i I_i (\pm\text{套连})$$

$$\oint_L \vec{B}\cdot d\vec{l} = \begin{cases} 0 \\ \pm\mu_0 I \end{cases}$$